Database Concepts - Assignment 4

Model Design and data Manipulation

(Task B and C)

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Task B

Objective:

Task B.1 aims to develop an Entity-Relationship (ER) diagram for a relational database capable of efficiently storing a provided dataset. The design must conform to recognized principles of effective database design, emphasizing normalization to reduce redundancy and uphold data integrity.

Aim:

The primary objective of this task is to utilize Lucidchart to create an ER diagram in UML notation. This diagram will illustrate a meticulously structured relational database schema tailored to accommodate the dataset.

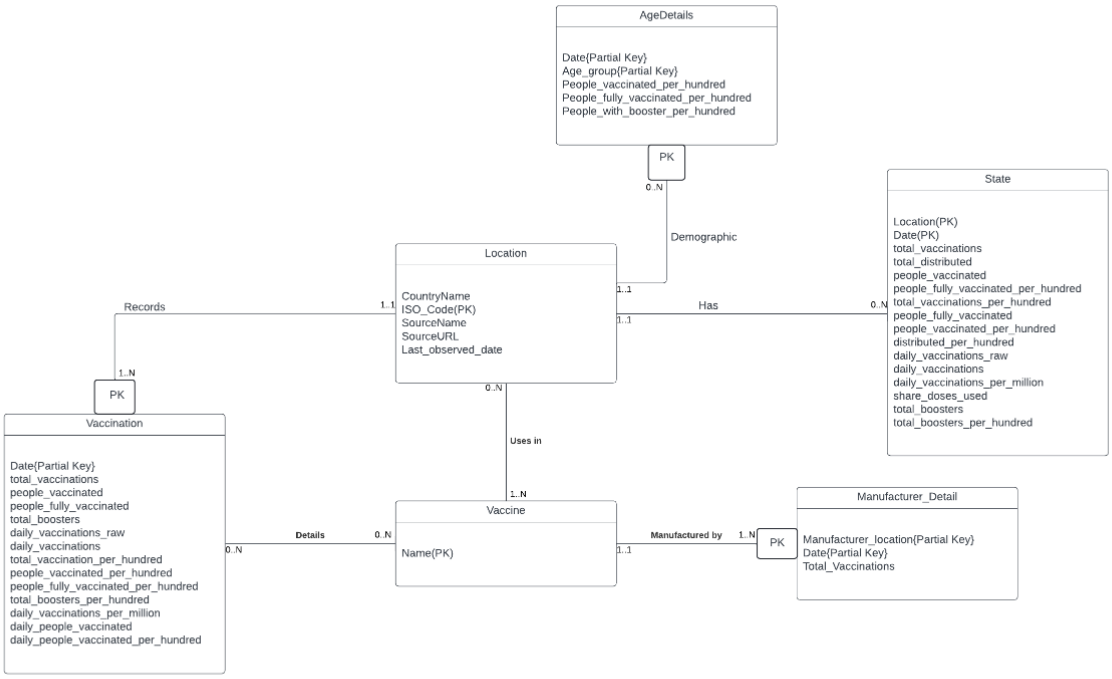
ER Diagram –

An Entity-Relationship (ER) diagram is a graphical tool used to design databases systematically and clearly. It illustrates entities (like objects or concepts), their attributes (properties), and the relationships that connect them within a database schema.

Each CSV was assumed to be an entity and then analysed. The multi-valued attribute was decomposed and made into a new entity.

Assumptions for the ER model:

1) In the future, states from other countries can be added to the database.
2) A vaccination may be undergoing tests, thus, not being used and represented in the records.
3) A vaccine may have 1 or more manufacturers.
4) A particular vaccine may or may not have manufacturer records.
5) A country may not have data records based on an individual's age.
6) A particular source name can provide data for multiple locations.
7) Data can be retrieved from multiple source locations and URLs for a particular country.
8) A particular vaccination record may be linked with more than one vaccine as the dataset depicts 'Vaccine' as a multi-valued attribute.
9) Not all locations are Vaccine manufacturers.



Mapping the ER model to a schema-

7 steps for mapping of ER Diagram to Database Schema-

1. **Map Strong Entities -**

Location (ISO_Code, Location, SourceURL, SourceName, Last_observation_date)

Vaccine (VaccineName)

State (Date, State, total_vaccinations, total_distributed, people_vaccinated, people_fully_vaccinated, people_fully_vaccinated_per_hundred, total_vaccinations_per_hundred, people_vaccinated_per_hundred, distributed_per_hundred, daily_vaccinations_raw, daily_vaccinations, daily_vaccinations_per_million, share_doses_used, total_boosters, total_boosters_per_hundred)

2. **Map Weak Entities -**

Manufacturer_Detail (Date, Location_Manufacturer, VaccineName*, TotalVaccination)

Age_Detail (Date, Age_group, ISO_Code*, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, people_with_booster_per_hundred)

Vaccination_Record (ISO_Code*, Date, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters, daily_vaccinations_raw, daily_vaccinations,

total_vaccinations_per_hundred, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, total_boosters_per_hundred, daily_vaccinations_per_million, daily_people_vaccinated, daily_vaccinated_per_hundred)

3. **Map 1:1 Relationships:**

No 1:1 relationship exists in the ER diagram.

4. **Map 1:N Relationships:**

State(Date, State, total_vaccinations, total_distributed, people_vaccinated, people_fully_vaccinated, people_fully_vaccinated_per_hundred, total_vaccinations_per_hundred, people_vaccinated_per_hundred, distributed_per_hundred, daily_vaccinations_raw, daily_vaccinations, daily_vaccinations_per_million, share_doses_used, total_boosters, total_boosters_per_hundred, ISO_Code*)

5. **Map M:N Relationships:**

Records (Date*, VaccineName*, ISO_Code*)

Uses_in (ISO_Code*, VaccineName*)

6. **Map Multi-valued Attributes:**

No multi-valued attributes exist in the ER diagram.

7. **Map Higher degree relationships:**

No ternary relationships exist in the ER diagram.

Final Schema:

Location (ISO_Code, Location, SourceURL, SourceName, Last_observation_date)

Vaccine (VaccineName)

State (Date, State, total_vaccinations, total_distributed, people_vaccinated, people_fully_vaccinated, people_fully_vaccinated_per_hundred, total_vaccinations_per_hundred, people_vaccinated_per_hundred, distributed_per_hundred, daily_vaccinations_raw, daily_vaccinations, daily_vaccinations_per_million, share_doses_used,total_boosters, total_boosters_per_hundred, ISO_Code*)

Manufacturer_Detail (Date , Location_Manufacturer, Vaccine_Name*, Total_Vaccination)

AgeDetails (Date, Age_group, ISO_Code*, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, people_with_booster_per_hundred)

Vaccination_Record (ISO_Code*, Date, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters, daily_vaccinations_raw, daily_vaccinations, total_vaccinations_per_hundred, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, total_boosters_per_hundred, daily_vaccinations_per_million, daily_people_vaccinated, daily_vaccinated_per_hundred)

Records (Date, VaccineName*, ISO_Code*)

Uses_in (ISO_Code*, VaccineName*)

Normalisation checks:

Normalization is crucial in organizing databases to prevent redundancy and ensure data reliability.

For the given database schema and ER model, the top-down validation approach will be useful. This approach helps ensure that the database structure follows the rules of normalization (1NF, 2NF, and 3NF).

To provide a brief, it will help us guarantee that the database that has been created is efficient and maintains data integrity.

Steps in the Top-Down Validation Approach:

1. **Understanding Functional Dependencies**:
    - Functional dependencies show how attributes relate.

In the vaccination database these are the FDs:

- **Location relation: Location** (ISO_Code, Location, SourceURL, SourceName, Last_observation_date)

  FD1: ISO_Code -> Location, SourceURL, SourceName, Last_observation_date

- **Vaccine relation: Vaccine** (VaccineName)

  Only has the primary key attribute. No FDs exist.

- **State Relation: State**(Date, State, total_vaccinations, total_distributed, people_vaccinated, people_fully_vaccinated, people_fully_vaccinated_per_hundred, total_vaccinations_per_hundred, people_vaccinated_per_hundred, distributed_per_hundred, daily_vaccinations_raw, daily_vaccinations, daily_vaccinations_per_million, share_doses_used,total_boosters, total_boosters_per_hundred, ISO_Code*)

FD1: Date, State -> total_vaccinations, total_distributed, people_vaccinated, people_fully_vaccinated, people_fully_vaccinated_per_hundred, total_vaccinations_per_hundred, people_vaccinated_per_hundred, distributed_per_hundred, daily_vaccinations_raw, daily_vaccinations, daily_vaccinations_per_million, share_doses_used,total_boosters, total_boosters_per_hundred, ISO_Code

- **Manufacturer_Detail Relation: Manufacturer_Detail** (Date , Location_Manufacturer, Vaccine_Name*, Total_Vaccination)

  FD1: Date, Location_Manufacturer, Vaccine_Name -> Total_Vaccination

- **AgeDetails Relation: AgeDetails**(Date, Age_group, ISO_Code*, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, people_with_booster_per_hundred)

  FD1: Date, Age_group, ISO_Code -> people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, people_with_booster_per_hundred

- **Vacciation_Record relation: Vaccination_Record** (ISO_Code*, Date, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters, daily_vaccinations_raw, daily_vaccinations, total_vaccinations_per_hundred, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, total_boosters_per_hundred, daily_vaccinations_per_million, daily_people_vaccinated, daily_vaccinated_per_hundred)

  FD1: ISO_Code, Date -> total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters, daily_vaccinations_raw, daily_vaccinations, total_vaccinations_per_hundred, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, total_boosters_per_hundred, daily_vaccinations_per_million, daily_people_vaccinated, daily_vaccinated_per_hundred

- **Records Relation: Records** (Date, VaccineName*, ISO_Code*)

  Only has the primary key attributes. No FDs exist.

- **Uses_in relation: Uses_in** (ISO_Code*, VaccineName*)

  Only has the primary key attribute. No FDs exist.

2. **Verification against Normalization Forms**:

   - **1NF (First Normal Form)**: The 1st Normal form makes sure that there are no comma separated values (multi-valued attributes) in the dataset and all attributes can be derived from the Primary key.

     In the database ER schema, there are all individual attributes and no multi-valued one, thus, the schema is in 1NF.

   - **2NF (Second Normal Form)**: Ensures that every non-key attribute is fully dependent on the primary key. This ensures no part of the primary key alone determines an attribute.

     In the database schema all elements are derived from the whole primary key and thus the schema is in 2NF

   - **3NF (Third Normal Form)**: This is to ensure that no transitive dependencies exist in the schema.

     A transitive relationship is when a non primary key attribute is dependent on another non primary key attribute.

     In the given schema there are no such dependencies and thus the model is in 3NF.

Task C

Cleaning and Manipulation-

Structure of the datasets doesn't match the ER model's and thus the data needed to be cleaned and manipulated.

Concepts learned in Data wrangling were used to pre-process the datasets.

1) The multi-valued attribute, Vaccine (in location and country datasets) was decomposed and made into a separate entity.
   This was done using excel's data tool to decompose the column and then R's pivot_longer function was used to make the manipulated CSV file.

2) Vaccine names were extracted from the Vaccines dataset and compared against the vaccine names from the manufacturer dataset. SKYCovione is only present in the latter.
   SKYCovione was added to the vaccine table using the INSERT command in SQLite.

3) The vaccination dataset had values with ISO_Code starting with OWID. However, in the locations dataset most of these values didn't exist. These were removed before importing the dataset.

   These values were as follows:

   'OWID_AFR', 'OWID_ASI', 'OWID_EUR', 'OWID_EUN', 'OWID_HIC', 'OWID_LIC', 'OWID_LMC', 'OWID_NAM', 'OWID_OCE', 'OWID_SAM', 'OWID_UMC', 'OWID_WRL'

4) While importing the dataset for the Uses_in relation, it was noted that for the country with ISO_Code BGD, the vaccine name 'Pfizer/BioNTech' was repeated, which gave the primary key constraint error. This was removed before the import.

5) Not all locations are mapped as the manufacturer. Only few of them are and thus Manufacturer location is a separate attribute in the Manufacturer_Detail entity table.

References

Dr Zhou, Xiangmin(Emily). (2024). https://rmit.instructure.com/courses/124806/modules. Canvas.

Lucid charts. https://www.lucidchart.com.